ABT RECERTIFICATION EXAMINATION #21

May 5, 2004 *PART A*

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 1995 and 2000. There are three parts in this exam and an answer sheet is provided for each part. These answer sheets must be returned to ABT, P.O. Box 30054, Raleigh, NC 27622 for grading. An analysis of your performance in each part of the exam will be provided to the recertification committee of ABT and they will contact you regarding this and other requirements of the recertification process. If you have any questions please contact the ABT Office at (919) 841-5022 or abtox@mindspring.com.

EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.

2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic and cardiovascular toxicology, hematopoietic toxicology, renal toxicology.

3. General principles, toxicokinetics, toxicodynamics, and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology.

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet ONLY on the line that says name.
5. Do not bend, cut, staple or tape answer sheets.

Good Luck!

Key for the following questions
Choose the single most appropriate answer

1. The major toxic effect of hydrogen cyanide exposure is
   A) lung damage
   B) hemoglobin alteration
   C) hemolysis of RBCs
   D) inhibition of mitochondrial respiration
   E) lipid peroxidation

2. Which of the following ionizing radiations has the shortest range (i.e., travels the shortest distance in tissue) for the same initial energy?
   A) alpha particle
   B) beta particle
   C) gamma ray
   D) x ray
   E) cosmic ray

3. Chemicals that alter the homeostasis of hormonal systems in the body have been termed "endocrine disruptors" or "endocrine modulators." Select the following chemical classes below to which this term applies.
   A) phthalates
   B) dioxins
   C) phytoestrogens
   D) organochlorines
   E) all of the above

4. Which of the following agents would NOT likely produce reactive airways dysfunction syndrome (RADS)?
   A) carbon monoxide
   B) chlorine
   C) ammonia
   D) toluene diisocyanate
   E) acetic acid

5. Which of the following metals is less toxic in its organic form compared to its inorganic form?
   A) mercury
   B) lead
   C) arsenic
   D) cadmium
   E) tin

6. Exposure to fumes of which of the following metals is most likely to cause acute chemical pneumonitis and pulmonary edema?
   A) lead
   B) zinc
   C) cadmium
   D) copper
   E) magnesium

7. An individual exposed to 10 rads (0.1 Gy) of whole body x-irradiation would be expected to
   A) have a severe bone marrow depression
   B) die
   C) be permanently sterilized
   D) exhibit no symptoms
   E) vomit

8. Which heavy metal is responsible for beer drinker's cardiomyopathy?
   A) cadmium
   B) lead
   C) copper
   D) cobalt
   E) chromium

9. Which of the following compounds would be expected to inhibit cholinesterase without oxidative metabolism?
   A) methyl parathion
   B) sumithion (fenitrothion)
   C) dichlorvos (DDVP)
   D) parathion
   E) dimethoate

10. Benzene is similar to toluene
   A) in its metabolism to redox active metabolites
   B) regarding covalent binding of its metabolites to proteins
   C) in its ability to produce CNS depression
   D) in its ability to produce acute myelogenous leukemia
   E) in its ability to be metabolized to benzoquinone

11. Which of the following is characteristic of asbestos fiber pneumoconiosis?
   A) "plastering" of the alveolar walls with asbestos fibers
   B) diffuse interstitial fibrosis
   C) accumulation of reticulin fibers, but not of collagen fibers
   D) nodular localized lesions of macrophages and asbestos fibers
   E) interstitial edema

12. The therapeutic efficacy of nitrite in the treatment of cyanide poisoning depends upon
   A) the formation of methemoglobin
   B) an increase in oxidative phosphorylation
   C) an increase in coronary arterial blood flow
   D) a decrease in intracranial pressure
   E) relaxation of the nonvascular smooth muscle

13. Consumption of milk from goats which have grazed on lupine plants containing the alkaloid, anagyrine, may cause
A) birth defects when ingested by women during early pregnancy
B) severe liver damage characterized by centrilobular necrosis
C) dizziness, nausea, headaches and hallucinations
D) numbness of the extremities
E) aphrodisia and a general increase in sexual awareness

14. Sorbitol and other sugar alcohols have been associated with
A) respiratory distress syndrome
B) osmotic diarrhea
C) hepatotoxicity
D) immediate hypersensitivity reaction
E) CNS depression

15. Aldrin, dieldrin and endrin are
A) dichlorodiphenylethanes
B) organophosphorous compounds
C) hexachlorocyclohexanes
D) cyclodienes
E) anticoagulants

16. Dichlorodiphenyltrichloroethane (DDT)
A) reduces potassium transport through nerve membranes
B) inhibits Na/K and Ca/Mg ATPases
C) exposure induces microsomal enzymes
D) all of the above
E) none of the above

Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

17. Which of the following statements is/are a correct pairing of a metal and its principal toxic effects?
1) lead - tremors
2) arsenic - anemia
3) mercuric chloride - renal failure
4) methyl mercury - demyelination of peripheral nerves

18. Ethanol is metabolized to acetaldehyde by
1) alcohol dehydrogenase
2) cytochrome P450 (CYP)2E1
3) catalase
4) squalene epoxidase

19. Inhalation of very large amounts of small inert particles, so-called nuisance dusts, into the deep lung can result in lung injury by which of the following mechanisms?
1) overwhelming the mucociliary escalator located in the trachea and bronchi, greatly reducing the efficiency of particle movement
2) causing toxic changes in the cilia of mucociliary escalator located in the trachea and bronchi and lessening the ability to move particles
3) increasing the viscosity of the mucus in the mucociliary escalator to alter both the ability to trap particles and the ability of cilia to move such particles
4) phagocytosis and migration of the pulmonary alveolar macrophages are overwhelmed by quantities of respirable dusts far in excess of physiological loads

20. Which of the following is/are true regarding solvents, gases and vapors?
1) a gas is a state of matter in which molecules are practically unrestricted by cohesive forces. A gas has no definite shape nor volume
2) a solvent is the constituent of a solution which is present in the larger amount; or the constituent which is liquid in the pure state, in the case of solutions of solids or gases in liquids
3) the term "vapor" and "gas" are often used interchangeably. Vapor is more frequently used for a substance, which though present in the gas phase, generally exists as a liquid or solid at room temperature. Gas is more frequently used for a substance that generally exists in the gaseous phase at room temperature
4) CCl_4 is a vapor; iodine and O_2 are gases

21. Glycols associated with renal toxicity include
1) propylene glycol
2) diethylene glycol
3) hexylene glycol
4) ethylene glycol

22. The effectiveness of sodium fluoroacetate (Compound 1080) as a rodenticide is attributed to which of the following?
1) it is well absorbed via oral ingestion
2) it is metabolized to fluorocitrate
3) the rapid metabolic rate in rodents
4) slow systemic release of fluoride ion

23. 1,1,2-Trichloroethylene
1) produces liver cancer in some strains of mice
2) produces renal toxicity in male rats
3) is present at many hazardous waste sites
4) does not produce CNS depression

24. Primates are more susceptible to methanol-induced acidosis and retinal lesions than rodents because
1) primates have higher hepatic levels of tetrahydrofolate
2) rodents use catalase to oxidize methanol
3) primates use alcohol dehydrogenase to oxidize methanol
4) rodents oxidize formate to CO_2 more rapidly

25. When considering the inhalation of fibers, which of the following statements is/are true?
1) both man-made and naturally occurring fibers are associated with lung injury
2) fiber composition, durability and aspect ratio are important to lung injury potential
3) fiber diameter is most important in determining fiber respirability
4) fiber shape has little influence on respirability

26. Mercury vapor (elemental mercury)
1) is primarily an occupational exposure hazard, although exposure may also occur from dental amalgam
2) is primarily an environmental exposure hazard, with exposure most likely to come from the consumption of seafood
3) may result in increased excitability, tremors, and gingivitis
4) is associated with paresthesia

27. The strobilurins are a relatively new class of fungicides used to protect food crops. They
1) inhibit the succinate dehydrogenase complex in the electron transport chain, thereby stopping the synthesis of aspartate and glutamate
2) are synthetic derivatives of a natural substance found in certain mushrooms
3) protect against fungal infection, but do not stop sporulation
4) inhibit mitochondrial respiration by stopping electron transfer between cytochrome b and cytochrome c

28. Which of the following statements regarding chlorophenoxy herbicides is/are correct?
1) cause chloracne
2) cause dermal irritation
3) biopersist and biomagnify in the environment
4) cause paresthesias and myotonia

29. The half-life of lead in bone is more than 200 times longer than the half-life in blood or soft tissue. In children, 70 % of the lead body burden is in bone. A 2-year-old child has an initial blood-lead level of 80 ug/dL. Immediately after chelation therapy, the blood level drops to 20 ug/dL, but is 45 ug/dL two months after chelation. Based on these data, which of the following is/are likely true?
1) chelation therapy is not effective in treating lead poisoning in children
2) the child may still be exposed to lead in his/her environment
3) the blood-lead level will return to 20 ug/dL in an additional two months
4) lead coming from the bones is contributing to the blood-lead level

30. Which is/are true about *in utero* and/or neonatal exposure to lead?
1) likely produces permanent cognitive defects
2) more readily absorbed through the gastrointestinal tract and retained than in adults
3) crosses the placenta
4) toxicity is enhanced by dietary deficiencies in calcium and iron

31. The sites of deposition of gases in the respiratory tract define the pattern of toxicity of those gases. Select the correct statement(s) below
1) ozone with high reactivity and low water solubility elicit toxic responses only in the nasopharyngeal region
2) relatively water insoluble gases such as NO_2 penetrate deeply into the lung, where they can elicit toxic responses
3) when SO_2 is inhaled with aerosols that can absorb the gas, its toxicity is reduced
4) H_2S efficiently passes through the respiratory tract because of very low water solubility, and can be taken up by the pulmonary blood supply

32. Vinclozolin is a dicarboximide fungicide. Which of the following is/are correct?
1) its metabolites can act as androgen antagonists
2) vinclozolin itself has no endocrine disrupting properties, but its metabolites are estrogenic
3) its metabolites exert their toxicity on the testis
4) it has been shown to cause egg shell thinning in exposed wildlife populations

33. Occupational exposure to toluene can produce which of the following effects in humans?
1) axonal degeneration
2) narcosis
3) leukemia
4) eye irritation

34. Which of the following statements about saxitoxin is/are correct?
1) it is produced by the dinoflagellate *Gonyaulax* during algal blooms known as "red tide"
2) it kills shellfish which accumulate the toxin by filter feeding
3) it blocks the influx of sodium ions in peripheral neurons without altering potassium or chloride conductances
4) it acts by inhibiting the Na^+/K^+ ATPase in peripheral neurons

35. Which of the following statements concerning marine toxins is/are true?
1) marine toxins can be classified according to their anatomical origin as icthyosarcotoxins (muscle, skin, liver or intestines), icthyootoxins (reproductive tissues), icthyohemotoxins (circulatory system), and icthyohepatotoxins (liver)
2) saxitoxin, the cause of paralytic shellfish poisoning (PSP), originates from plankton and produces paresthesias and neuromuscular weakness
3) domoic acid, the cause of amnesic shellfish poisoning (ASP), is an analogue of kainic acid, originates primarily in diatoms and algae, and affects primarily the central nervous system
4) tetrodotoxin, the cause of puffer fish poisoning, originates primarily in endosymbiotic bacteria and is an hepatotoxin producing fatty liver

36. Which of the following is/are considered to be a stochastic effect of radiation exposure?
1) gastrointestinal tract damage
2) cataracts of the eye lens
3) sterility
4) hereditary effects

37. Which of the following concerning linear energy transfer (LET) is/are true?
1) LET is much higher for gamma rays than for alpha and beta particles
2) high LET radiation has a high energy loss per unit path length
3) high LET radiation generally has a low ionization density
4) a high LET is generally associated with particles with a relatively large mass

38. Which of the following plants is/are capable of causing malformations in offspring when the plants are ingested during pregnancy?
1) *Veratrum californicum* (false hellbore)
2) *Lupinus* species (lupine)
3) *Conium maculatum* (water hemlock)
4) *Manihot esculenta* (cassava)

39. Which of the following factors have/has a significant effect on the toxicity of one or more metals?
1) formation of metal-lipid complexes
2) age and stage of development of the exposed individual
3) body weight of exposed individuals of the same age
4) lifestyle factors (diet, environmental exposures) of the exposed individual

40. Which of the following plant materials is/are responsible for microtubule-associated neurotoxicity?
1) colchicine
2) paclitaxel
3) vincristine
4) hexachlorophene

41. Regarding the plant toxin ricin,
1) early manifestations of toxicity include nausea, vomiting, and diarrhea
2) its mechanism of toxicity is the inactivation of the ribosomal structure, thereby blocking protein synthesis
3) it is composed of two peptide chains linked by a disulfide bond
4) it is found in the nut of both the Ohio buckeye and horse chestnut tree

42. Which of the following, when ingested, produce conduction defects and bradycardia?
1) foxglove (*Digitalis purpurea*)
2) monkshood (*Aconitum* spp.)
3) oleander (*Nerium oleander*)
4) mistletoe (*Phoradendron flavescens*)

43. Hypertension and elevated LDL cholesterol are associated with abuse of which of the following agents?
1) gamma-aminobutyric acid (GABA)
2) cocaine
3) amphetamines
4) anabolic steroids

44. Which of the following statements on direct or indirect food additives is/are correct?
1) aspartame is a non-nutritive sweetener that induces a dose-dependent stimulation of catecholamine biosynthesis by phenylalanine that results in CNS effects including headache, dizziness, and seizures
2) FD and C Red dye No. 3 induces thyroid tumors in rats by altered thyroid hormone synthesis and hyperstimulation of the thyroid. It is unlikely to be a human health risk since humans are much less sensitive to elevated TSH levels than are rats and the human exposures are very low
3) fungal contaminants, zearalenone and zearalenol, interact with the androgen receptors, induce biphasic changes in luteinizing hormone, usually accumulate in edible tissues and contaminate milk and meats
4) fungal contaminants in grain such as ergot alkaloids are smooth muscle stimulants that promote vasoconstriction and uterine contractions, antagonize serotonin, and block the CNS responses to epinephrine

45. The term "delayed neurotoxicity" of organophosphates refers to which of the following?
1) the time needed after uptake of an indirect acting organophosphate for metabolic activation
2) the continuous uptake of small amounts of organophosphate with a slowly progressing inhibition of cholinesterases eventually leading to symptoms of toxicity
3) the persistence of some of these agents on soil and vegetable matter which can still cause serious intoxication in man and livestock many weeks after application of the organophosphate
4) axonal degeneration and demyelination caused by certain organophosphates days or weeks after a single exposure

46. Which of the following statements is/are true regarding the toxicity of glycol ethers?
1) 2-methoxyethanol is toxic to the testis and developing fetus
2) humans are less susceptible than rats to glycol ether-induced hemolysis
3) reproductive toxicity is produced by the alkoxy acid metabolite of 2-ethoxyethanol
4) ethylene- and propylene-based glycol ethers have similar toxicity and potency characteristics

47. Hypolipidemic agents derived from fibric acid have been demonstrated to
1) lower cholesterol
2) act as peroxisome proliferators in rodents
3) cause an increase in hepatic tumors after long-term administration to rats and mice
4) lead to liver tumors in humans

48. Which of the following agents can induce hemolytic anemia in persons with glucose-6-phosphate dehydrogenase deficiency?
1) copper
2) sulfisoxazole
3) primaquine
4) thiourea

49. Which of the following is/are true about butylated hydroxytoluene (BHT)?
1) BHT is an antioxidant food additive
2) BHT may act as a free radical trapping agent
3) at high chronic doses BHT induces cytochrome P450 enzymes that metabolize estrogen and thyroxine
4) at high chronic doses BHT induces thyroid tumors in rats

50. The potential to induce *torsades de pointe* (a potentially lethal cardiac arrhythmia) is increased by drugs that
1) block delayed rectifier potassium channels
2) slow AV node conduction (increase PR interval)
3) increase cardiac action potential duration (prolong QTinterval)
4) block the inward current of calcium ions

ABT RECERTIFICATION EXAMINATION #21

May 5, 2004 *PART B*

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 1995 and 2000. There are three parts in this exam and an answer sheet is provided for each part. These answer sheets must be returned to ABT, P.O. Box 30054, Raleigh, NC 27622 for grading. An analysis of your performance in each part of the exam will be provided to the recertification committee of ABT and they will contact you regarding this and other requirements of the recertification process. If you have any questions please contact the ABT Office at (919) 841-5022 or abtox@mindspring.com.

EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.

2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic and cardiovascular toxicology, hematopoietic toxicology, renal toxicology.

3. General principles, toxicokinetics, toxicodynamics, and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology.

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet ONLY on the line that says name.
5. Do not bend, cut, staple or tape answer sheets.

Good Luck!

Key for the following questions
Choose the single most appropriate answer

1. The purpose of a photocarcinogenicity study is to determine if a test substance
   A) is activated to a carcinogen when exposed to ultraviolet radiation
   B) is a secondary carcinogen
   C) is a mutagen
   D) promotes tumor production and/or the earlier appearance of tumor formation following exposure to ultraviolet radiation which is the primary carcinogen
   E) none of the above

2. The first nephron segment involved in urinary concentration is the
   A) proximal tubule
   B) descending limb of Henle's loop
   C) ascending limb of Henle's loop
   D) distal convoluted tubule
   E) pars recta

3. Which of the following leads to deposition of oxalate crystals in the urinary tract?
   A) *Dieffenbachia* (dumbcane)
   B) propylene glycol
   C) methoxyflurane
   D) philodendron
   E) ethylene glycol

4. Which of the following is NOT considered immunotoxic?
   A) diethylstilbestrol (DES)
   B) cyclophosphamide
   C) dichlorodiphenyltrichloroethane (DDT)
   D) vinyl ether
   E) toluene diisocyanate

5. A target organ of streptomycin-induced toxicity is
   A) lung
   B) heart
   C) reproductive system
   D) kidney
   E) liver

6. Which of the following is characteristic of a nongenotoxic carcinogen?
   A) has no influence on the promotional stage of carcinogenesis
   B) would be expected to produce positive responses in *in vitro* assays for mutagenic potential
   C) typically exerts other forms of toxicity and/or disrupts cellular homeostasis
   D) generally shows little structural diversity
   E) typically has little effect on cell turnover

7. A single large dose of N-nitrosodimethylamine fails to induce cancer in rats, but repeated dosing induces cancer because
   A) the single large dose is lethal, while the threshold for cancer induction can be exceeded by repeated smaller doses
   B) the main DNA lesion from a single large dose can be repaired readily by methyltransferase, while repeated smaller doses can deplete the available repair enzyme, induce mutations in DNA, and effectively induce cancer
   C) the enzyme system involved in detoxification of N-nitrosodimethylamine is depleted after repeated doses, allowing N-nitrosodimethylamine to build up and exceed the threshold for cancer induction
   D) the enzyme system involved in conversion of N-nitrosodimethylamine to the active carcinogen is induced and on subsequent repeated doses more active carcinogen is produced
   E) the initial dose of N-nitrosodimethylamine causes cell damage and, thus, high mitotic rates; subsequent small doses induce mutations in DNA and effectively induce cancer

8. Which of the following is NOT a neurotransmitter in the central nervous system?
   A) somatostatin
   B) gamma-aminobutyric acid
   C) glutamic acid
   D) glycine
   E) kainic acid

9. Based on the available data, cyclosporin A-induced cholestasis is most likely explained by which of the following?
   A) interaction with the cytoplasmic receptor, cyclophilin
   B) inhibition of transcription of message (mRNA) for critical cytokines
   C) induction of programmed cell death in centrilobular hepatocytes
   D) inhibition of ATP-dependent transporter(s) in the bile canalicular membrane
   E) depolarization of the mitochondrial membrane potential

10. Alkylating agents and antimetabolites induce bone marrow damage primarily characterized by which of the following?
   A) pancytopenia
   B) aplastic anemia
   C) thrombocytopenia
   D) granulocytopenia
   E) none of the above

11. In regard to chemically-induced adverse effects on the eye,
   A) no chemical has been shown to cause glaucoma
   B) nonionic detergents damage the eye more that cationic detergents
   C) 2,4-dinitrophenol, corticosteroids and naphthalene are known to cause cataracts in humans
   D) methanol produces blindness by rendering the cornea and lens opaque
   E) acids usually produce late-appearing ocular toxicity as contrasted to alkalis which produce immediate damage

12. Allergic contact dermatitis is
   A) a non-immune response caused by a direct action of an agent on the skin
   B) an immediate type I hypersensitivity reaction
   C) a delayed type IV hypersensitivity reaction
   D) characterized by the intensity of reaction being proportional to the elicitation dose
   E) not involved in photo-allergic reactions

Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

13. Which of the following pairings of drugs and alterations in human development is/are correct?
   1) thalidomide and limb malformations
   2) 13-cis-retinoic acid (Accutane®) and ear, heart, brain malformations
   3) ethanol and craniofacial dysmorphisms
   4) Bendectin®and microcephaly

14. Which of the following statements is/are correct regarding the *Salmonella* mutagenicity assay (Ames test) with S9 metabolic activation?
   1) the tester strains contain several mutations which increase their sensitivity to mutagens
   2) the tester strains synthesize histidine, but lose this ability upon exposure to mutagens
   3) the tester strains will detect chemicals which induce both base-pair substitutions or frameshift mutations
   4) all mutagens detected as positive in this test system require metabolism to the ultimate mutagenic compound via an exogenous mammalian metabolic activation system

15. Sympathomimetic action of cocaine results from
   1) binding to beta adrenergic receptors
   2) stimulating the uptake of catecholamines
   3) increased synthesis of catecholamines
   4) blocking reuptake of catecholamines

16. Which of the following substances readily cross(es) the mature, intact blood/brain barrier?
   1) mannitol
   2) D-glucose
   3) unconjugated bilirubin
   4) 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP)

17. Caloric restriction, as opposed to *ad libitum* feeding, in rodents improves longevity, lowers incidence and severity of chronic, age-associated disease and results in delayed onset of spontaneous tumors. Potential mechanisms include
   1) reduction in age-associated oxidative injury and/or glycation
   2) increased physical activity level
   3) decreased cell proliferation rates
   4) reduction in intake of dietary carcinogens

18. Characteristics of stem cell spermatogonia include which of the following?
   1) relative resistance to a variety of insults
   2) more susceptible to toxic injury than proliferative spermatogonia
   3) their destruction leads to irreversible loss of spermatogenic capacity
   4) divide by meiosis and thereby contribute to genetic diversity

19. The ICH Guidelines for detection of toxicity to reproduction introduced a reduced treatment interval of four weeks for males before pairing. This reduction was based on which of the following consideration(s) by ICH?
   1) recent studies have shown that the length of the spermatogenesis cycle in the male rat is only 25 days and not 60 days, as previously thought
   2) histopathology of the testis has been shown to be the most sensitive method for the detection of effects on spermatogenesis
   3) many studies have shown that treatment of adult males for longer than 4 weeks may lead to non-treatment related stress and non-specific damage to the testicular tissue
   4) compounds affecting spermatogenesis almost invariably affect postmeiotic stages, and a 4-week treatment period is sufficient to detect these changes

20. Which of the following is/are true regarding transgenic animal assays?
1) some transgenic mouse models contain a shuttle vector with bacterial genes as mutational targets
2) p53 transgenic mouse models are used because of the frequency of p53 mutations in many human cancers
3) transgenic animals may be used to study the induction of tissue-specific mutations
4) all transgenic animal models have an extra DNA fragment in their genome

21. Clastogenesis is
1) typically detected with the Ames test
2) typically a terminal event and, thus, is usually cell lethal
3) a leading cause of teratogenesis
4) often related more to progression of cancer rather than cancer initiation

22. Myelin is the affected target in the CNS following exposure to
1) triethyltin
2) cuprizone
3) hexachlorophene
4) trimethyltin

23. Bronchoconstriction and chronic pulmonary disease can result from inhalation of materials that appear to act primarily through an allergic response. The underlying mechanism is related to circulating or fixed antibodies to specific components of the inhaled materials. Which of the following combinations of diseases is/are examples of allergic responses?
1) farmers' lung, bagassosis, byssinosis
2) farmers' lung, bagassosis, silicosis
3) bagassosis, byssinosis, mushroom pickers' lung
4) bagassosis, byssinosis, talcosis

24. Which of the following is/are NOT evaluated in a typical teratology (embryo-fetal development or Segment II) study?
1) ossification of skull bones
2) fetal weight following vaginal delivery
3) atrioventricular septum of the heart
4) male fertility

25. Which of the following contribute(s) to the toxicity of dust?
1) particle size
2) concentration of the dust
3) exposure duration
4) the chemical form of the dust

26. The major factor(s) that distinguish upper respiratory tract irritants from lower respiratory tract irritants is/are
1) molecular weight
2) water solubility
3) potency as oxidizing agents
4) initial irritant properties

27. The European Committee for the Validation of Alternative Methods (ECVAM) conducted a multiple laboratory validation test to develop *in vitro* models for embryo toxicity testing . The *in vitro* assays that were validated were
1) the rat embryo limb bud micromass assay
2) the mouse embryonic stem cell test
3) the rat embryo culture test
4) the Hershberger assay

28. In a Type I hypersensitivity reaction, the chemical mediators that are released can produce
1) vasodilation with local edema
2) increased vascular permeability
3) constriction of bronchial smooth muscle
4) cytotoxic hemolysis

29. Vinca alkaloids (vincristine and vinblastine) and colchicine
1) produce an axonopathy
2) are agents which cause primary damage to the perikarya of peripheral neurons
3) damage elements of microtubule-based transport as part of their mechanism of action
4) act indirectly on the nervous system via primary toxic effects in the liver

30. Data required under U.S. EPA's health effects test guideline for developmental neurotoxicity include(s)
1) motor activity
2) time of vaginal opening
3) auditory startle
4) thymus weights

31. The pulmonary toxicity of paraquat is
1) similar to that of diquat
2) expressed after oral exposure
3) only observed after chronic exposure
4) characterized by fibrosis

32. Which of the following contribute(s) to the development of fatty liver?
1) increased fatty acid oxidation
2) decreased apoprotein synthesis
3) increased synthesis of low density lipoproteins
4) decreased secretion of low density lipoproteins

33. The plaque-forming cell (PFC) response to sheep red blood cells (SRBC) is used to assess substances for immunotoxic effects. Which of the following statements is/are true?
1) the assay measures the functional capability of the humoral immune response
2) the PFC response to SRBC is a monitor of the primary effector function of the B lymphocyte
3) the cooperation of multiple cell types is necessary for the PFC response in this assay
4) the assay can be used by itself to ascertain the mechanism of immunosuppression

34. Which of the following agent:lesion pairs is/are correct regarding the liver?
1) alpha-naphthylisothiocyanate: cholangiodestructive cholestasis
2) copper: periportal hepatocyte necrosis
3) aflatoxin B_1: hepatoma
4) valproic acid: fatty liver

35. Hepatic responses to chronic ethanol abuse are characterized by
1) impaired diffusion of material from the sinusoids to the space of Disse
2) lipid accumulation caused by impaired ability to synthesize lipoproteins
3) induction of cytochrome P450 (CYP) 2E1 in acinar Zone 3
4) progressively decreased capacity for detoxifying products formed by the catabolism of amino acids and hemoglobin

36. Which of the following immune system responses is/are pro-inflammatory?
1) complement activation
2) innate immunity
3) Type I hypersensitivity
4) humoral immunity

37. Which of the following is/are correct?
1) as a general rule, irritant dermatitis reactions can be elicited from a much lower quantity of agent than is required for a skin sensitization response
2) the inflammatory responses from exposures to known skin irritants and skin allergens can include erythema and induration
3) photoallergy and phototoxicity require induction and elicitation reactions
4) photoallergic and phototoxic skin reactions may be produced by systemic or topical exposure to the agent

38. Glutathione has which of the following roles in protecting the lens of the eye from chemically-induced toxicity?
1) maintenance of protein sulfhydryls in the reduced form
2) protection from oxidative damage
3) conjugation of hydrophobic compounds having electrophilic centers
4) source of the amino acids glutamate, cysteine and glycine

39. Respiratory allergy or atopy is associated with the development of an immunological response characterized by IgE secretion. This is due, in part, to the differentiation of uncommitted CD4 T-helper cells into the effector T-helper-2 (Th2) pathway. Which cytokine(s) has/have been implicated in the development of this pathway?
1) interleukin-5
2) interferon-gamma
3) interleukin-4
4) tumor necrosis factor

40. Which of the following is/are correct concerning chloracne?
1) it is an indication of significant exposure to 2,3,7,8-tetrachlorodibenzo-*p*-dioxin
2) humans are unique among the primates for susceptibility to chloracne following exposure to chlorinated aromatic hydrocarbons
3) mild cases of chloracne clear up spontaneously within a few months after cessation of exposure, while severe cases persist for as long as 30 years
4) the skin of the anterior upper torso is usually involved first in the expression of chloracne

41. Amphotericin B produces a nephrotoxic response characterized by which of the following?
1) renal tubular acidosis
2) impaired renal concentrating ability
3) hypokalemia
4) uremia

42. Xenobiotics may cause cancer by which of the following mechanisms?
1) metabolic activation to a highly reactive electron-deficient species
2) formation of oxygen or nitrogen radical species
3) interaction with specific receptor proteins to produce ligand-protein complexes that are capable of modifying the expression of DNA
4) enhancement of cellular proliferation

43. In the male reproductive tract
1) small molecular weight molecules (e.g., water, urea) can readily cross the blood-testis barrier
2) the mammalian gonad is incapable of metabolizing foreign chemicals that have traversed the blood-testis barrier
3) 2-methoxyethanol affects all stages of spermatogenesis and some stages of spermiogenesis
4) the Sertoli cell is the primary site of testosterone synthesis

44. According to ICH Guidelines, a study of "Fertility and Early Embryonic Development to Implantation"
1) detects effects on the estrous cycle, tubal transport, implantation, and development of preimplantation stages of the embryo in treated female animals
2) is usually conducted in one species, preferably rats
3) generally employs a pretreatment of males for 4 weeks prior to mating, unless data from other studies suggest that this should be modified
4) allows detection of effects on male reproductive function (i.e., libido, epididymal sperm maturation)

45. Nongenotoxic carcinogens may cause neoplasia by
1) stimulating mitogenesis
2) inhibiting apoptosis
3) stimulating expression of mutated proto-oncogenes
4) inducing apoptosis

46. Which of the following metals is/are associated with hepatotoxicity?
1) titanium
2) zinc
3) platinum
4) copper

47. Occupational inhalation exposure to which of the following fiber types has been shown to be strongly associated with mesotheliomas and bronchogenic carcinomas?
1) rock wool fibers
2) man-made vitreous fibers
3) glass fibers
4) asbestosis fibers

48. The principal determinant(s) of renal accumulation of nephrotoxic compounds include which of the following?
1) high blood flow to the kidney
2) high concentrations of metabolizing enzymes in the kidney
3) active transport systems in renal tubules
4) rate of urinary excretion of high molecular weight proteins

49. Which is/are true regarding bromobenzene?
1) metabolized to both ortho- and para-bromophenol
2) nephrotoxic but not hepatotoxic
3) has very low toxicity *in vitro* when added to renal cortical slices
4) hepatotoxic but not nephrotoxic

50. The most effective treatment of cyanide poisoning includes
1) sodium nitrite
2) sodium thiosulfate
3) oxygen
4) penicillamine

ABT RECERTIFICATION EXAMINATION #21

May 5, 2004

PART C

This examination constitutes part of the recertification process for toxicologists certified by the American Board of Toxicology in 1995 and 2000. There are three parts in this exam and an answer sheet is provided for each part. These answer sheets must be returned to ABT, P.O. Box 30054, Raleigh, NC 27622 for grading. An analysis of your performance in each part of the exam will be provided to the recertification committee of ABT and they will contact you regarding this and other requirements of the recertification process. If you have any questions please contact the ABT Office at (919) 841-5022 or abtox@mindspring.com.

EXAMINATION CATEGORIES

1. Metals, solvents, pesticides, gases, dusts and aerosols, industrial chemicals and radiation, plant and animal toxins, food additives, drugs and cosmetics.

2. Mutagenesis, carcinogenesis, developmental and reproductive toxicology, pulmonary toxicology, neurologic toxicology, immunologic toxicology, cutaneous toxicology, ocular toxicology, hepatic and cardiovascular toxicology, hematopoietic toxicology, renal toxicology.

3. General principles, toxicokinetics, toxicodynamics, and mechanisms of action, risk assessment, epidemiology and biostatistics, regulatory toxicology and risk management, environmental toxicology, occupational toxicology, forensic and clinical toxicology.

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet ONLY on the line that says name.
5. Do not bend, cut, staple or tape answer sheets.

Good Luck!

Key for the following questions
Choose the single most appropriate answer

1. A threshold limit value-time weighted average for a chemical represents
   A) an airborne concentration of a chemical that can never be exceeded
   B) an airborne concentration of a chemical that is believed to cause no adverse health effect to a worker exposed for 8 hours a day, 40 hours a week
   C) an airborne concentration of a chemical that cannot be exceeded for longer than 15 minutes a day
   D) a value for an acceptable airborne concentration of a chemical established by the Occupational Safety and Health Administration
   E) an airborne concentration which cannot be measured using available technology

2. The reference dose (RfD) is generally determined by applying which of the following default procedures?
   A) an uncertainty factor of 100 is applied to the NOAEL in chronic animal studies
   B) a risk factor of 1000 is applied to the NOAEL in chronic animal studies
   C) a risk factor of 10,000 is applied to the NOAEL in subchronic animal studies
   D) an uncertainty factor between 10,000 and 1 million is applied to the NOEL from chronic animal studies
   E) multiplying the NOAEL from chronic animal studies by 100

3. A virtually safe dose represents
   A) a maximum likelihood estimate of a safe exposure level
   B) an estimate of the dose that would result in no excess risk of cancer
   C) a dose corresponding to a 95 percent upper confidence limit on adverse response rates
   D) a reference dose for carcinogen exposure that uses an uncertainty factor of 1,000,000
   E) the dose associated with an excess risk of 1 in 1,000,000

4. Cofactor-dependent drug metabolism will generally occur in
   A) the soluble supernatant fraction (100,000 g for 90 minutes)
   B) the supernatant remaining after mitochondria are spun down
   C) lysosomes
   D) the purified nuclear fraction
   E) none of the above

5. Which of the following concerning the use of the "benchmark dose" in risk assessment is NOT correct?
   A) can use the full range of doses and responses studied
   B) allows use of data obtained from experiments where a clear "no observed adverse effect level" (NOAEL) has been attained
   C) may be defined as the lower confidence limit on the 10% effective dose
   D) is primarily used for analyses of carcinogenicity data and has limited utility for analyses of developmental and reproduction studies that generate quantal data
   E) is not limited to the values of the administered doses

6. The Hershberger assay involves exposing castrated adult male rats to compounds and weighing the accessory sex organs 5 to 7 days later. This test is an assay for
   A) estrogens
   B) anti-androgens and androgens
   C) goiterogens
   D) carcinogens
   E) mutagens

7. An epidemiologic study of lithium in drinking water as a cause of cardiac arrhythmias would NOT involve which of the following?
   A) identification of an exposed group
   B) a before-exposure to after-exposure comparison
   C) electrocardiographic measurements
   D) monitoring of lithium intake
   E) tobacco smoking history

8. Which of the following is the target molecule with which diptheria toxin interacts to cause its cellular toxicity?
   A) actin
   B) glutathione
   C) elongation factor 2
   D) synaptobrevin
   E) protein kinase C

9. Administration by oral gavage of a test compound that is highly metabolized by the liver versus subcutaneous injection will most likely result in
   A) less parent compound present in the systemic circulation
   B) more local irritation at the site of administration caused by the compound
   C) lower levels of metabolites in the systemic circulation
   D) more systemic toxicity
   E) less systemic toxicity

Key for the following questions
Choose A) if 1,2 and 3 are correct
Choose B) if 1 and 3 are correct
Choose C) if 2 and 4 are correct
Choose D) if only 4 is correct
Choose E) if all four answers are correct

10. Chemical agents pass through membranes by which of the following mechanisms?
1) facilitated diffusion
2) passive diffusion
3) active transport
4) filtration

11. Regarding dose-response distributions and their uses, which of the following is/are true?
1) a quantal response may be a graded response
2) logarithmic transformation of a cumulative normal distribution response will create a sigmoidal dose-response curve
3) the ratio of the LD50 to the ED50 (for a selected effect) can be a measure of the "margin of safety" for a pharmaceutical
4) a response that is not normally distributed because of a disproportionate number of highly-sensitive test animals will extend farther to the right of the mean dose than to the left of the mean dose

12. Factors which can significantly influence the passage of agents through the skin include
1) metabolism by epidermal tissue
2) plasma protein binding
3) solubility of the agent
4) evaporation from dermis

13. Which of the following inhibit(s) the function of target molecules?
1) morphine
2) strychnine
3) glutamate
4) tetrodotoxin

14. Which of the following pairs of toxicant-excretory routes is/are true?
1) ethanol - feces
2) methylmercury - biliary
3) beryllium - exhaled air
4) N-methylnicotinamide - urinary

15. The biological half-life is
1) a function of clearance
2) based on the elimination component of the concentration versus time curve
3) used to design repeated dosing regimens to obtain steady state blood levels
4) helpful in predicting accumulation

16. Which of the following is/are true?
1) Phase I reactions, such as oxidation, hydrolysis, and reduction, introduce or expose a functional group
2) Phase II reactions, such as glucuronidation and sulfation, result in increased xenobiotic hydrophilicity and greatly promote excretion
3) xenobiotic metabolizing enzymes are located in several subcellular compartments incuding lysosomes, cytosol, and mitochondria
4) Phase II biotransformation must always be preceded by Phase I

17. Which of the following is/are correct regarding the gastrointestinal tract?
1) weak bases are more rapidly absorbed from the small intestine than the stomach
2) weak acids are likely to be absorbed more rapidly from the large intestine than the stomach
3) substances absorbed from the intestine may enter the lymphatic system or the portal circulatory system
4) during oral administration, most drugs are extensively absorbed in the mouth

18. Conversion of doses between species is frequently based on body surface area and not on body weight. Which of the following is/are true?
1) body surface area is approximately proportional to (body weight) $^{2/3}$
2) body surface area is proportional to body clearance and volume of distribution
3) cancer chemotherapeutic drugs are usually administered to patients based on body surface area
4) the ratio of body surface area to body weight is the same for mice and humans

19. Which of the following accurately contrast(s) necrosis and apoptosis?
1) cell swelling and rupture versus cell shrinkage and phagocytosis
2) internucleosomal DNA fragmentation versus random DNA fragmentation by lysosomal enzymes
3) independent of genetic regulation versus dependent on genetic regulation
4) can be observed in all organs versus only observed in kidney and liver

20. Which of the following chemicals cause(s) hypoxia as a direct effect?
1) cyanide
2) pentachlorophenol
3) rotenone
4) carbon monoxide

21. Cytochrome P450 enzymes are
1) a family of related enzymes important in the metabolism of chemicals
2) enzymes that catalyze the oxidation of chemicals
3) enzymes that catalyze the reduction of chemicals
4) enzymes that occur in a wide variety of tissues and organisms

22. Which of the following is/are correct?
1) the heme iron in cytochrome P450 enzymes is usually in the ferric (Fe^{+3}) state when it binds ligands such as oxygen and carbon monoxide
2) cytochrome P450 enzymes catalyze the synthesis and metabolism of steroid hormones as well as the activation and detoxification of xenobiotics
3) cytochrome P450 enzymes are found exclusively in the microsomal fractions of liver cells
4) one basic reaction catalyzed by cytochrome P450 enzymes is monooxygenation in which one atom of oxygen is incorporated into a substrate and the second oxygen atom is reduced to water

23. Characteristics of apoptosis include
1) interception of the process leading to neoplasia if apoptosis occurs in precancerous cells
2) mitochondrial release of cytochrome c and nuclear DNA fragmentation
3) externalization of phosphatidylserine on plasma membrane
4) possible organ atrophy or dysfunction in nonregenerative tissues

24. Which of the following about toxicogenomic studies using microarrays is/are NOT true?
1) mRNA is extracted from cells and reacted with reverse transcriptase to produce labeled cDNA probes
2) the technique can simultaneously analyze thousands of genes
3) microarray tests may become more sensitive indicators of toxicity than animal tests
4) the technique can be used to test for gene induction, but not for gene suppression

25. Differential susceptibility to an adverse drug response
1) can be due to genetic variability
2) can be an induced response from a second drug
3) influences the strategy for treatment with certain anticancer agents
4) affects the dose that can be safely administered on an individual to individual basis

26. Which of the following activate(s) the $GABA_A$ receptor?
1) picrotoxin
2) barbiturates
3) cyclodiene insecticides
4) benzodiazepines

27. Compound X has both carcinogenic and non-carcinogenic effects. The RfD for oral exposure is 10 ug/kg/day. The slope factor (SF) is .0001 $(ug/kg/day)^{-1}$. The concentration in drinking water is 100 ug/L. Assuming that a person drinks 2 liters of water per day, weighs 70 kg, and is exposed for 70 years,
1) there is no concern for non-carcinogenic risks
2) there is concern for the carcinogenic risks
3) the risk for carcinogenesis is greater than the risk for non-carcinogenic effects
4) there is concern for non-carcinogenic risks

28. The microsomal enzyme cytochrome P450 (CYP) 1A1
1) is induced by cigarette smoke
2) catalyzes the hydroxylation and epoxidation of benzo[a]pyrene
3) is quantified by its ability to catalyze the O-dealkylation of 7-methoxyresorufin and 7-ethoxyresorufin
4) is expressed in the human liver

29. Which of the following reaction(s) is/are catalyzed by Phase I enzymes?
1) acylation
2) hydrolysis
3) methylation
4) reduction

30. Conjugation with glutathione is important in
1) detoxification of aflatoxin B_1
2) toxification of hexachlorobutadiene
3) excretion of methylmercury and arsenic
4) formation of haloalkane mercapturic acids

31. Which of the following statements regarding Generally Recognized As Safe (GRAS) substances according to the U.S. Food, Drug and Cosmetic Act is/are true?
1) GRAS substances are those which have a toxicologic data base sufficient to demonstrate they are "safe" as presently used
2) no new substances may be added to the GRAS list, but substances can be rescinded from the list
3) new substances may be added to the GRAS list, but listed substances cannot be rescinded
4) the FDA is required to review the safety of existing GRAS substances

32. For the risk assessment of carcinogens, the
1) slope in the low-dose linear region of the dose-response curve, sometimes referred to as Q1*, represents a potency estimate for carcinogens
2) slope in the experimental region of the dose-response curve, sometimes referred to as Q1*, represents a potency estimate for carcinogens
3) slope in the low-dose linear region of the dose-response curve, multiplied by the dose, estimates the probability of contracting cancer
4) area under the curve of the line defining the dose-response relationship is commonly used to represent a potency estimate for carcinogens

33. Hydrogen sulfide (sewer gas) poisoning is NOT characterized by
1) usually resulting from occupational exposure
2) symptoms similar to those of cyanide poisoning
3) production of conjunctivitis
4) being easily treated by administration of oxygen

34. Which statement(s) characterize(s) the use of biological monitoring to estimate occupational exposure to chemicals?
1) may be confounded by metabolic variations due to genetic polymorphisms
2) when involving the quantitation of a specific compound or metabolite, is typically dependent upon analyses obtained during measured or controlled human exposures
3) can provide important adjuncts to epidemiology studies
4) most commonly involves the analysis of hair

35. Which of the following tissues is/are collected for post-mortem analysis when metal toxicity is suspected?
1) bone
2) adipose tissue
3) kidney
4) ovary

36. Which of the following is important for evaluation of the potential hazards from occupational exposure to a chemical?
1) data from animal toxicology studies
2) pharmacokinetic parameters in animals and humans
3) route of likely exposure
4) molecular weight

37. Which of the following is/are associated with asbestosis?
1) pulmonary fibrosis
2) mesothelioma
3) pleural plaques
4) emphysema

38. Which of the following statements is NOT true regarding the Animal Welfare Act?
1) it requires an Institutional Animal Care and Use Committee
2) is administered through a section of the United States Department of Agriculture
3) requires assurances that alternatives to painful procedures were considered in the design of a study
4) covers the humane handling, care and transportation of any species used in research

39. Under the U.S. Toxic Substances Control Act of 1976, which establishes a broad, nationwide program for the federal regulation of the manufacture and distribution of toxic substances, the administrator of the EPA
1) is charged to compile and publish a list of all chemical substances manufactured in the US
2) may not prohibit the manufacture and distribution of a new chemical simply because toxicity data for this substance are not available
3) may impose restriction on the manufacture and distribution of a chemical substance which presents unreasonable risk of injury to health or the environment
4) may require any substance manufactured to be tested for establishing its toxicity

40. Jake the retriever brought home a full box of rat poison. When his owner found him having eaten a substantial amount of the contents, both Jake and the box were rushed immediately to the veterinarian. The active ingredient in the rat poison was found to be brodifacoum. What course of therapy was instituted?
1) induction of vomiting
2) repeated administration of vitamin K
3) checking blood clotting time after the treatment period
4) small injected doses of heparin

41. Heat shock proteins (HSP)
1) are both the signal that activates transcription of the HSP genes and the substrate for the gene product
2) have been used with varying degree of success as biomarkers, but require further validation before they can be used in risk assessment
3) are highly conserved and have been found in organisms as diverse as bacteria, mollusks, and humans
4) show highly transient elevation upon continuous exposure to a stressor

42. Which of the following has/have been found to be carcinogenic through human epidemiological studies?
1) arsenic
2) acetylene
3) coal tars
4) lead

43. Transformation of chemicals in/on the soil occur(s) by which of the following?
1) direct chemical reaction
2) photochemical reaction
3) biochemical reaction
4) bioaccumulation

44. Which of the following define(s) the first pass effect?
1) poor absorption due to rapid GI transit time
2) biotransformation by the gastrointestinal cells
3) elimination in the bile
4) clearance of a chemical before entry to the post hepatic systemic circulation

45. Which of the following is/are true about apparent volume of distribution?
1) it is the theoretical tissue volume in which a chemical is dissolved
2) it is a theoretical volume and not an exact biologically definable volume
3) it is defined as amount of chemical (dose) divided by its concentration in the body
4) it is a proportionality constant relating the body burden (or dose) of a substance to its plasma concentration

46. Which of the following is/are true for acrylamide?
1) exposure usually occurs by inhalation
2) chronic exposure results in polyneuritis in man
3) only sensory axons are affected and not motor neurons
4) only the monomer can cause peripheral neuropathy

47. Which of the following statements regarding ecological risk assessments is/are true?
1) ecological risk assessments are less complex and less controversial than most human risk assessments because they deal primarily with a single chemical entity
2) ecological effects are often more difficult to interpret than human health effects because they involve predictions about changes in populations and communities of organisms
3) indirect effects, such as changes in feed availability, habitat structure, or predator abundance, of the environmental toxin being studied seldom need to be considered because they represent such a small percentage of the overall effect of the toxin
4) statistical analysis of ecological risk often resorts to a worst case scenario due to the high degree of uncertainty in the risk assessment

48. Tributyltin, an anti-fouling agent used on the hulls of boats,
1) produces imposex which is the irreversible development of male characteristics in female mollusks
2) results in morphological changes in the reproductive tracts of mollusks, including formation of a penis and testes in females
3) yields populations of mollusks that are dominated by males
4) is particularly toxic to the abductor muscle

49. A study is being designed to test whether the time to respond to a tone is affected by the consumption of alcohol. X is the mean time to respond after consuming alcohol, and y is the mean time to respond before consuming alcohol. Which of the following is/are true?
1) the null hypothesis is there will be no effect of alcohol consumption on response time
2) the null hypothesis is $x - y = 0$
3) if the experimental data show no effect of alcohol consumption on response time, when indeed there is a real effect, this would be considered a Type I error
4) the null hypothesis would be rejected if it is shown that $x = y$

50. As part of a biological monitoring program, trichloroacetic acid has been identified in urine samples collected from workers who use solvents in a degreasing operation. The workers may have been exposed to
1) trichloroethylene
2) perchloroethylene (tetrachloroethylene)
3) methylchloroform (1,1,1-trichloroethane)
4) dichloromethane (methylene chloride)

ABT Answers

2004 Recent answer key - not authorized

Key #	Part A - Key	Part B - Key	Part C - Key
1	D	D	B
2	A	B	A
3	E	E	C
4	A	D	B
5	C	D	D
6	C	C	B
7	D	D	B
8	D	E	C
9	C	D	A
10	C	D	E
11	B	C	C
12	A	C	B
13	A	A	C
14	B	B	C
15	D	D	E
16	D	C	A
17	A	B	B
18	A	B	B
19	D	C	B
20	A	A	D
21	C	D	E
22	A	A	C
23	A	B	E
24	D	C	D
25	A	E	E
26	B	C	C
27	C	A	A
28	E	A	A
29	C	B	C
30	E	A	E
31	C	C	D
32	B	C	B
33	C	A	D
34	B	E	B
35	A	E	B
36	D	B	A
37	C	C	A
38	A	A	D
39	C	B	E
40	A	B	A
41	A	E	E
42	E	E	B
43	D	B	A
44	C	E	D
45	D	A	C
46	A	D	C
47	A	D	C
48	A	A	A
49	A	B	A
50	B	A	A